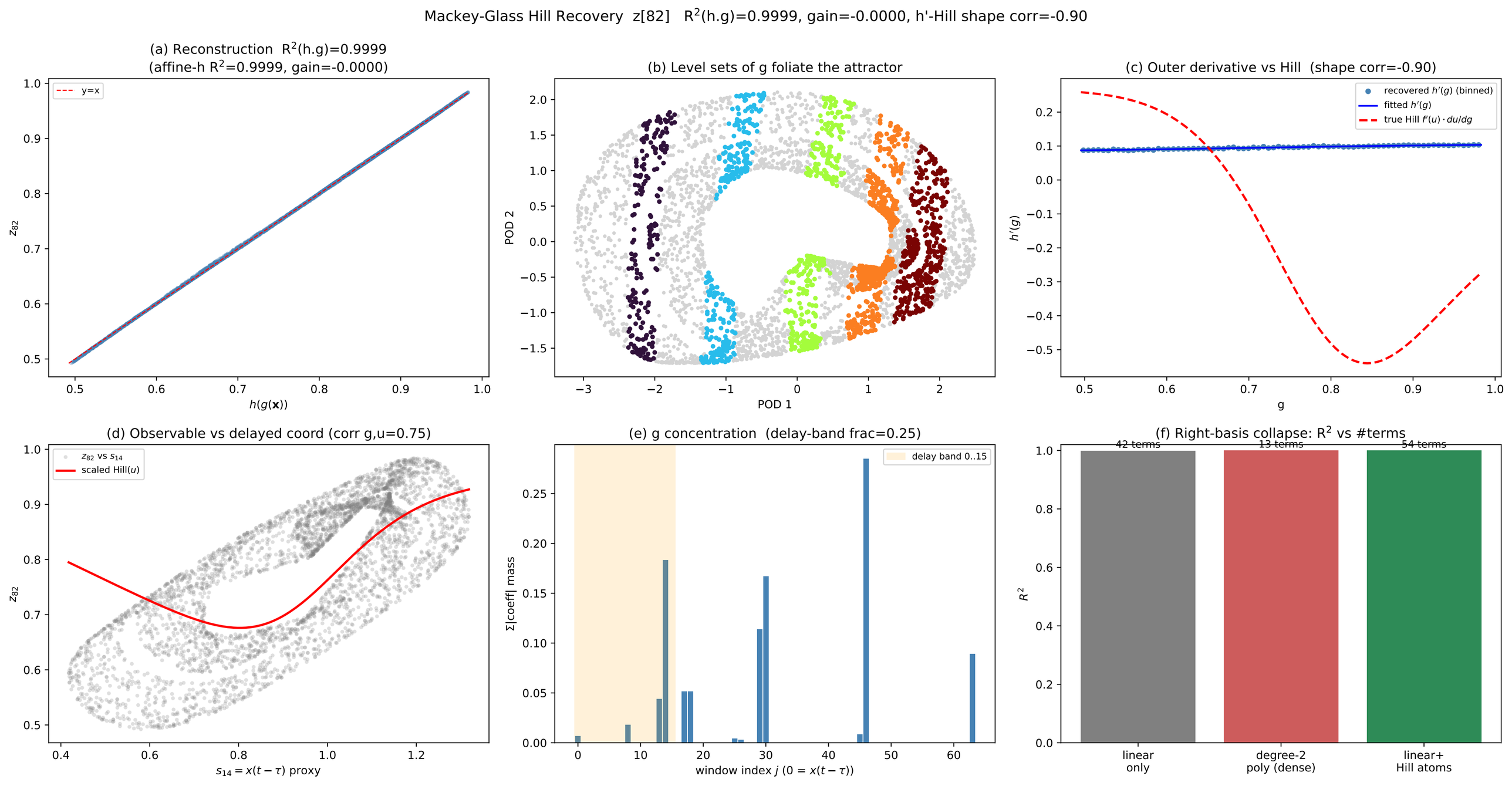

Mackey-Glass Hill Recovery z[82] R²(h.g)=0.9999, gain=-0.0000, h'-Hill shape corr=-0.90
(a) Reconstruction R²(h.g)=0.9999
(affine-h R²=0.9999, gain=-0.0000)
y=x
z_{82}
$h(g(\mathbf{x}))$
(b) Level sets of g foliate the attractor
POD 2
POD 1
(c) Outer derivative vs Hill (shape corr=-0.90)
recovered $h'(g)$ (binned)
fitted $h'(g)$
true Hill $f'(u)\cdot du/dg$
$h'(g)$
g
(d) Observable vs delayed coord (corr g,u=0.75)
z_{82} vs s_{14}
scaled Hill(u)
z_{82}
$s_{14} = x(t-\tau)$ proxy
(e) g concentration (delay-band frac=0.25)
delay band 0..15
$\Sigma|\text{coeff}|$ mass
window index j ($0 = x(t-\tau)$)
(f) Right-basis collapse: R² vs #terms
42 terms
13 terms
54 terms
R^2
linear
only
degree-2
poly (dense)
linear+
Hill atoms